

Translations

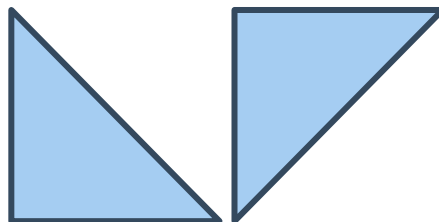


1 State the type of transformation that has occurred on the following shapes:

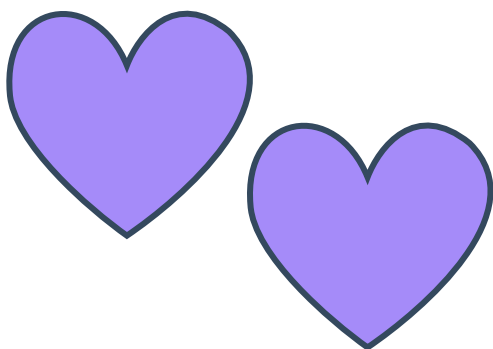
a



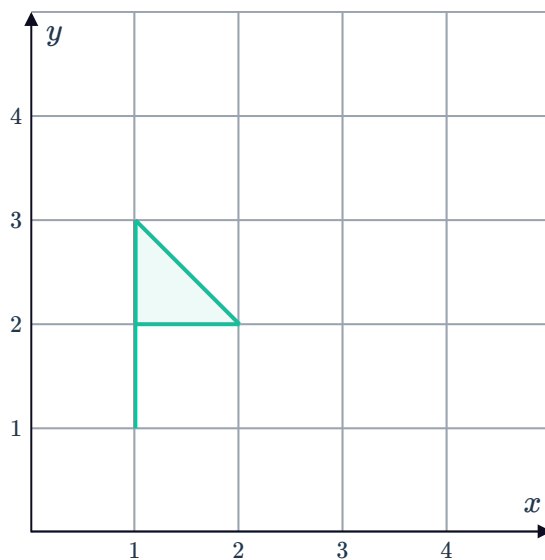
b



c

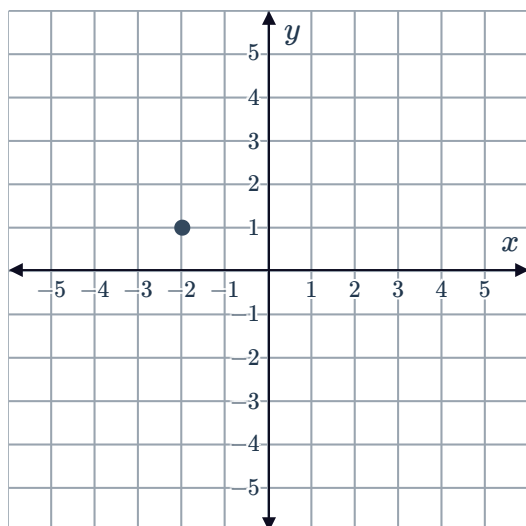


2 Translate the given flag 1 unit up and 2 units to the right on the number plane:

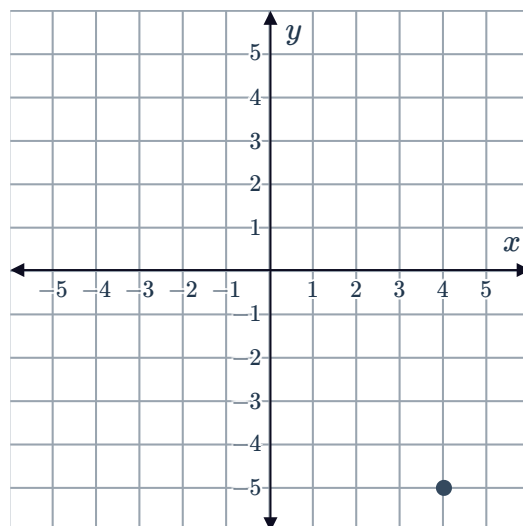


3 For each of the following points, plot the translated point using the given instruction:

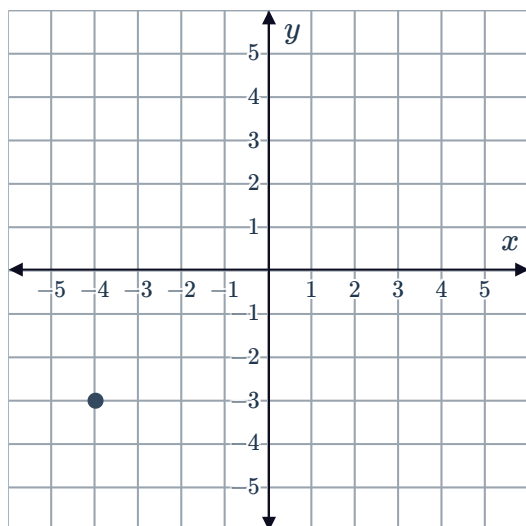
a Translate the point 3 units to the right.



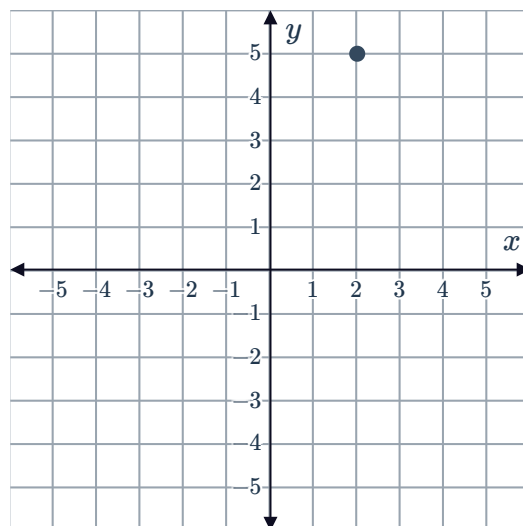
b Translate the point 4 units to the left.



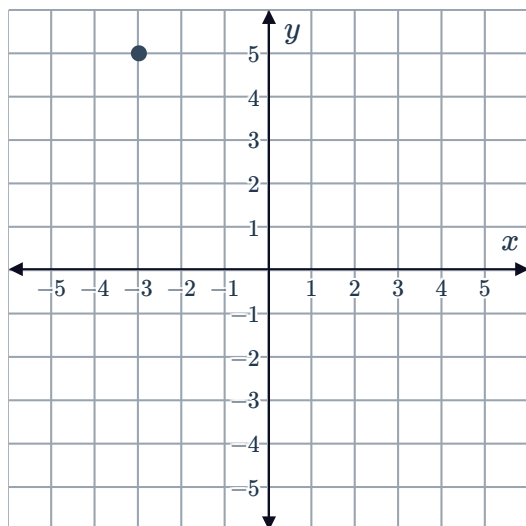
c Translate the point 7 units up.



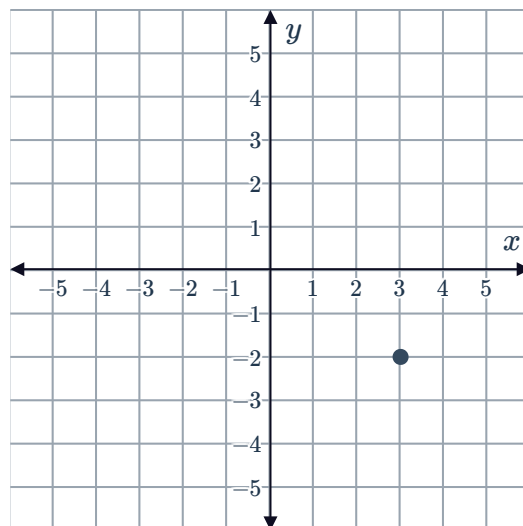
d Translate the point 4 units down.



e Translate the point 2 units to the right and 4 units down.



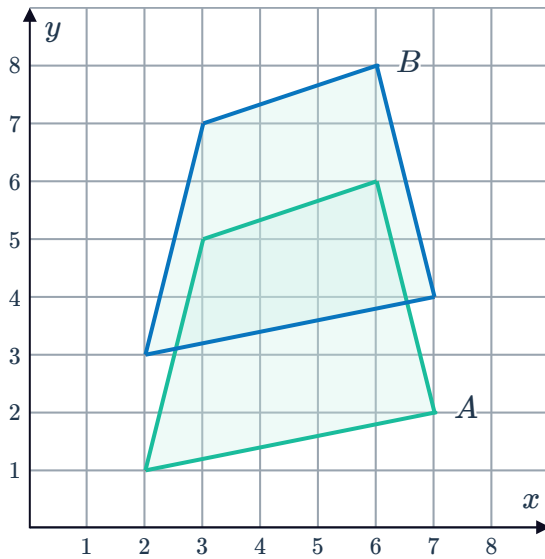
f Translate the point 5 units to the left and 6 units up.



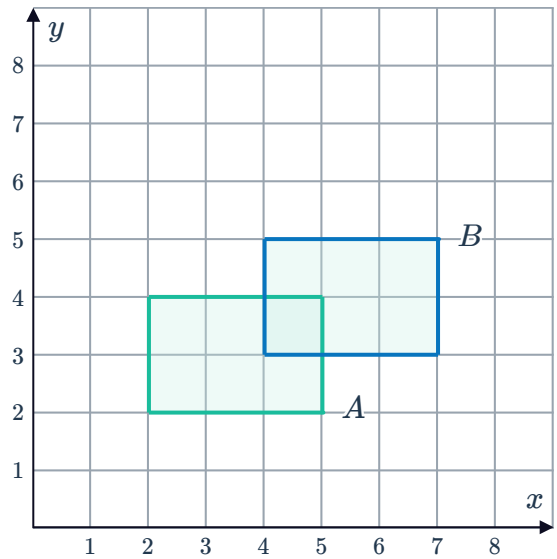
4 For each of the following, describe the transformation from figure *A* to figure *B*:



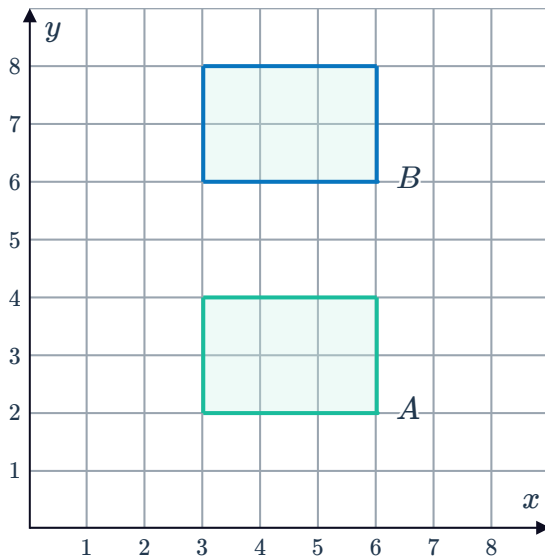
a



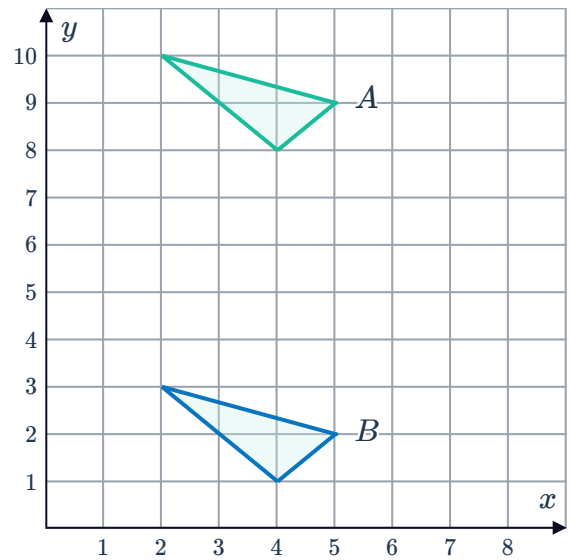
b



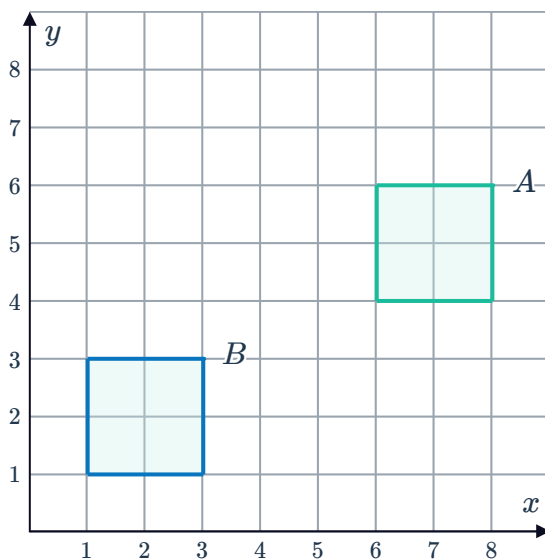
c



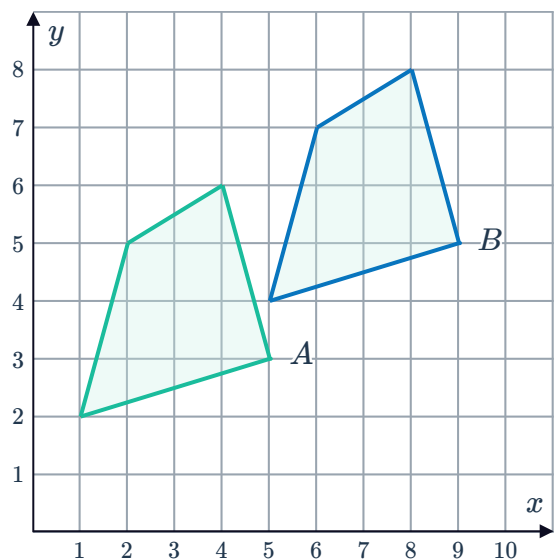
d



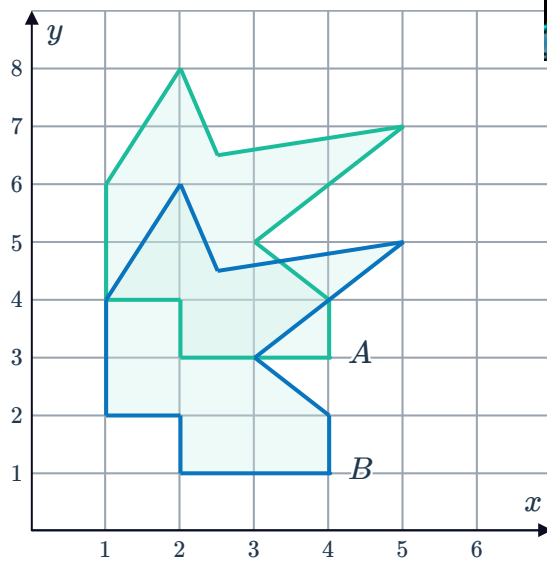
e



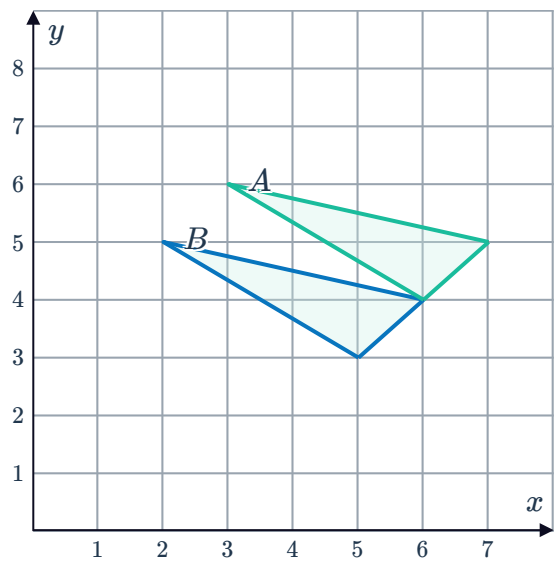
f



g

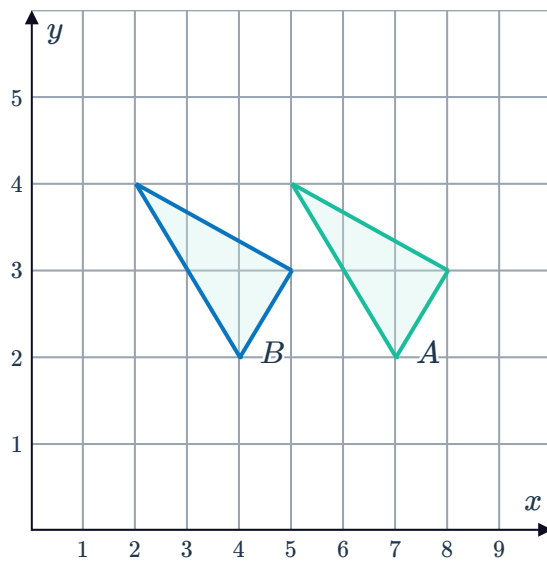


h

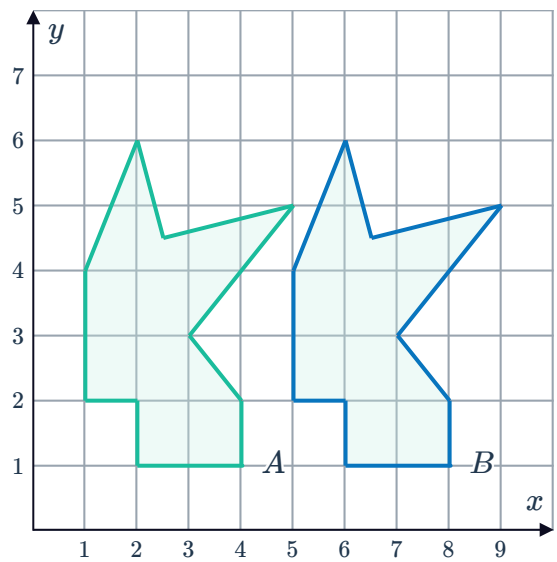


5 For each of the following, describe the translation from figure *B* to figure *A*:

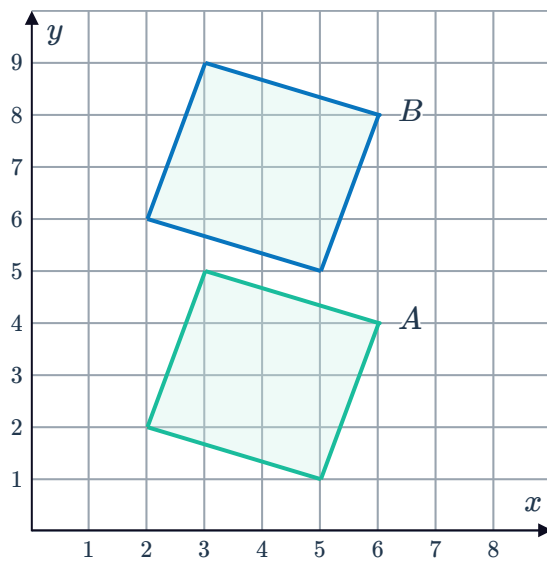
a



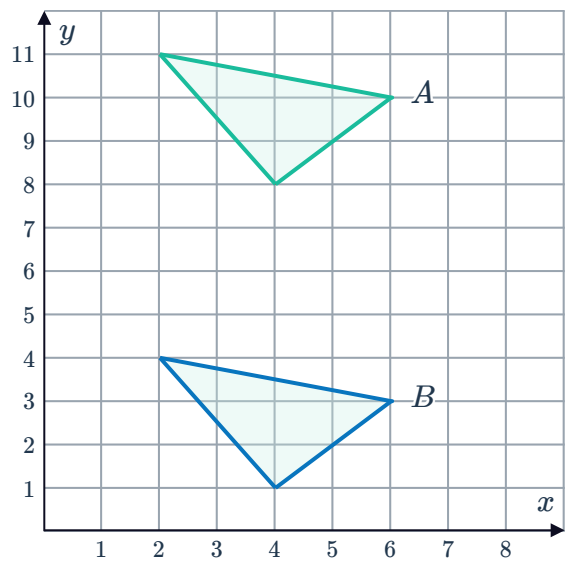
b



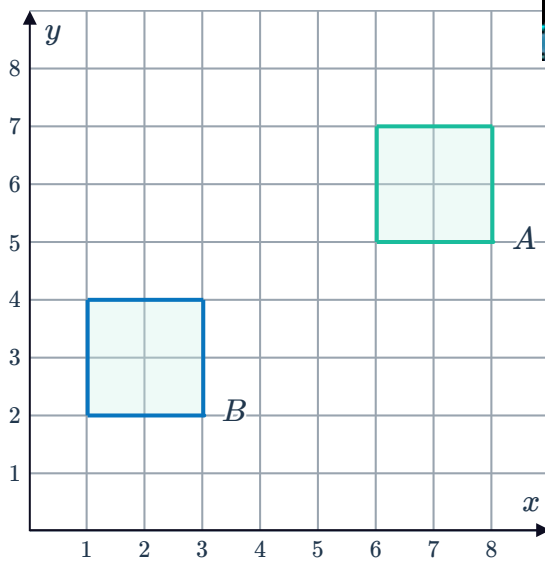
c



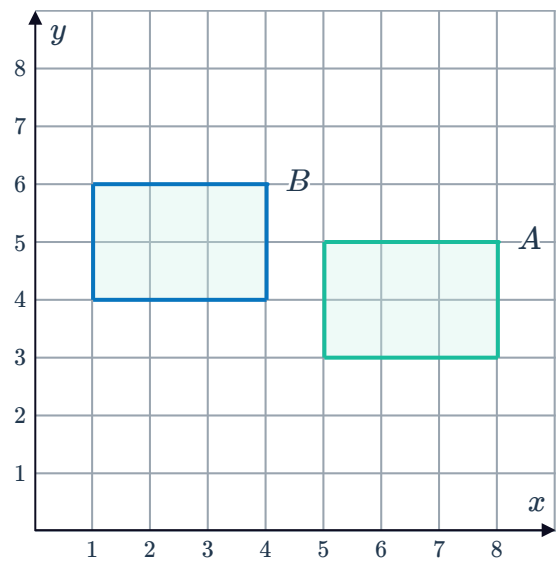
d



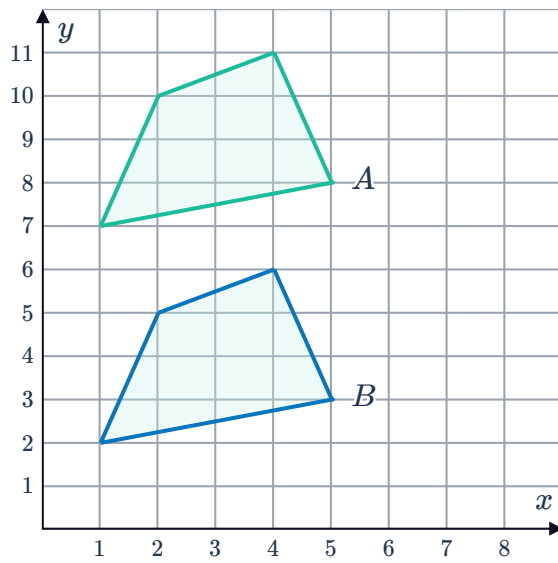
e



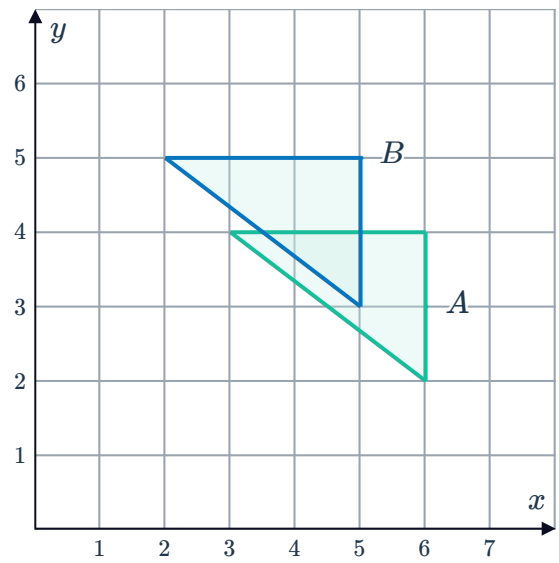
f



g

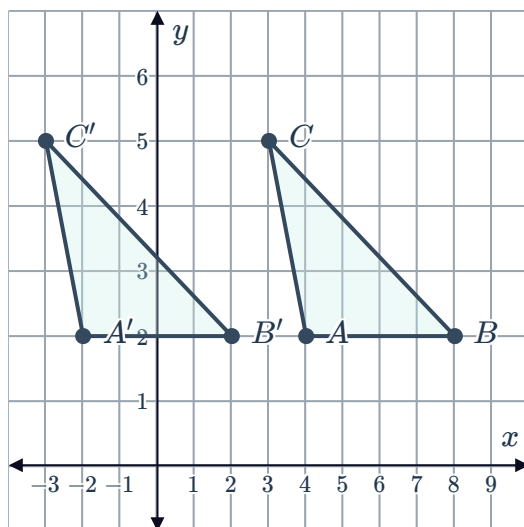


h

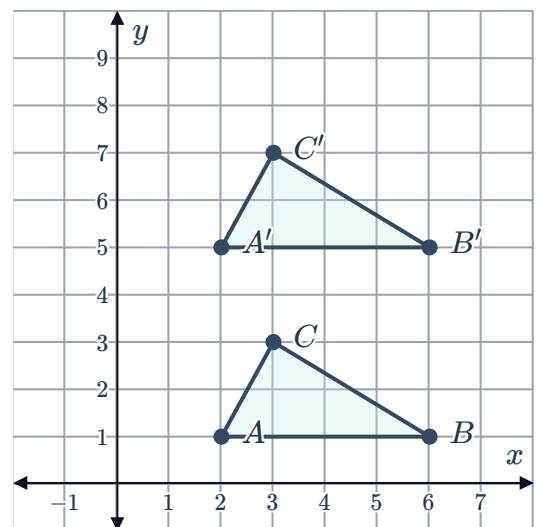


6 For each of the following, describe the translation of triangle ABC to triangle $A'B'C'$:

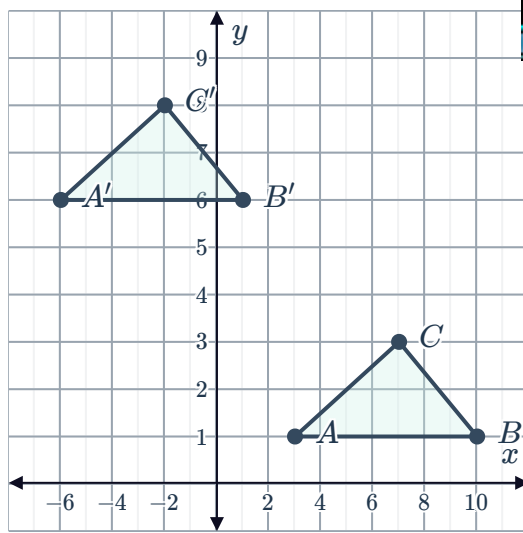
a



b

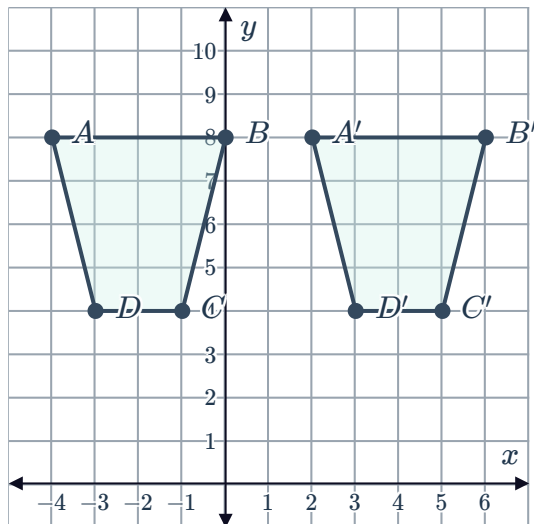


c

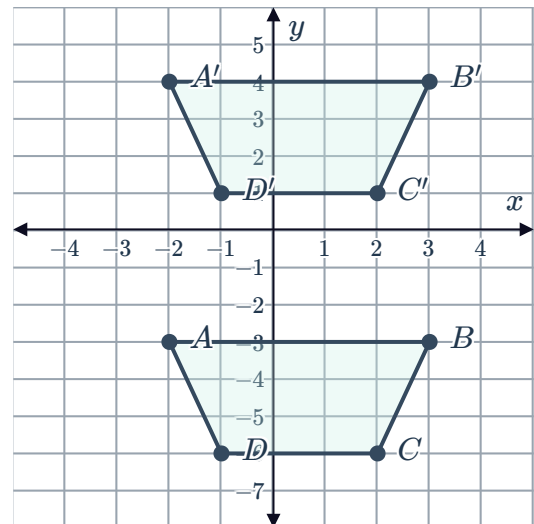


7 For each of the following, describe the translation of the trapezium $ABCD$ to the trapezium $A'B'C'D'$:

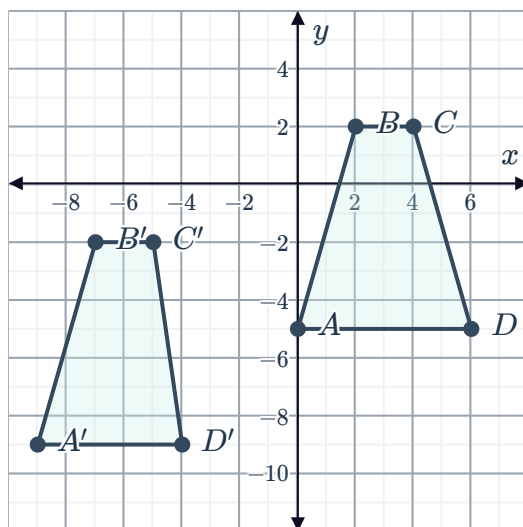
a



b



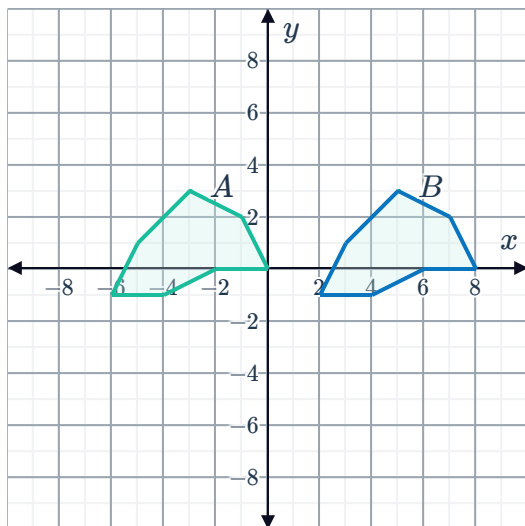
c



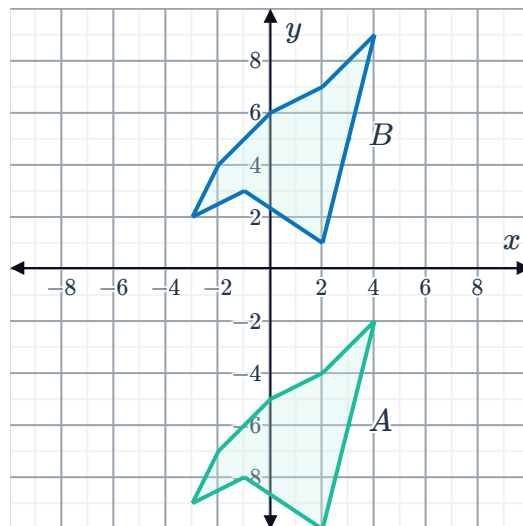
8 For each of the following, describe the translation of the polygon A to the polygon B :



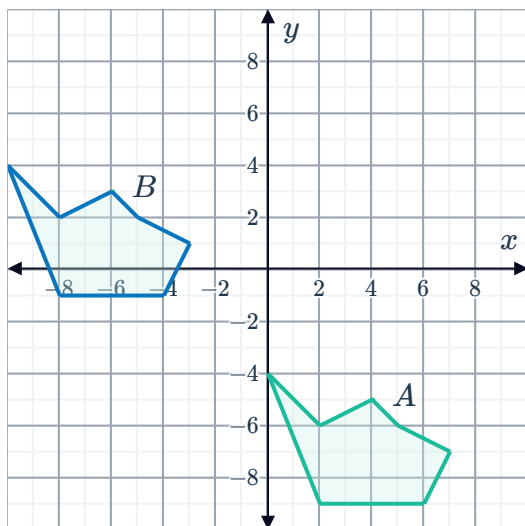
a



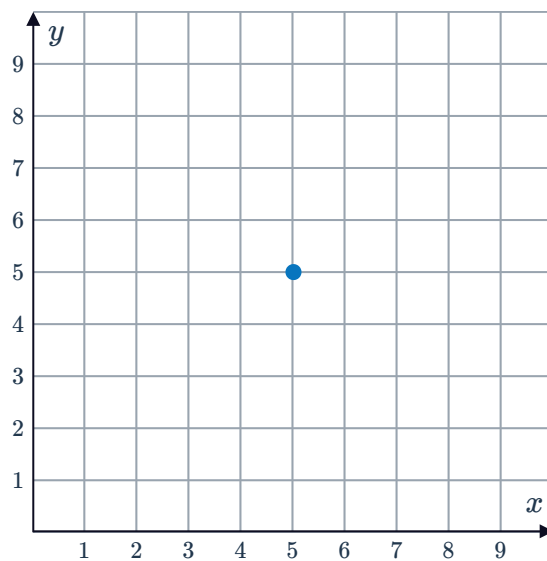
b



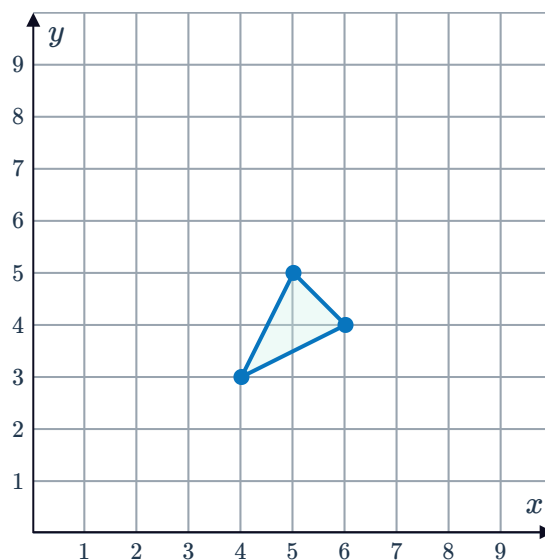
c



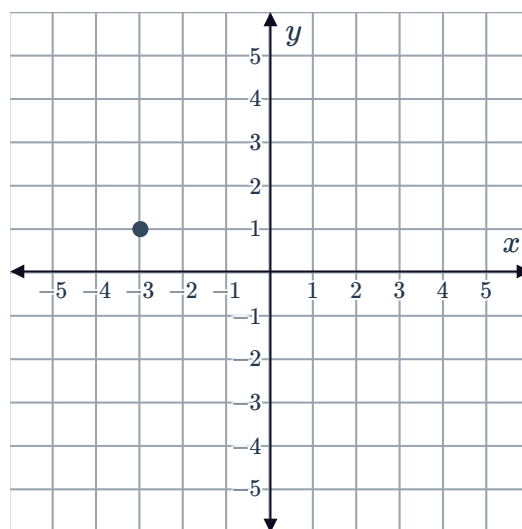
9 A point has been translated 2 units left. The translated point is shown on the right. Plot the starting position of this point before it was translated.



- 10 A triangle has been translated 3 units left. The translated triangle is shown on the right. Plot the starting position of the triangle before it was translated.



- 11 The point on the number plane is translated 4 units up. Write down the coordinates of its new position.



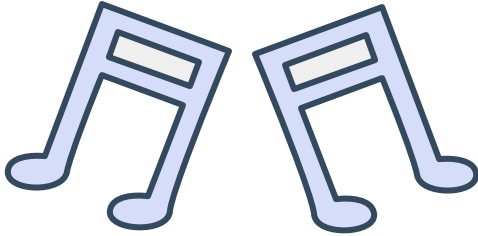
- 12 Point A is translated 3 units down and 4 units to the right, where it sits at point $(5, -2)$. Write down the coordinate of the original point A .



Reflections

13 State the type of transformation that has occurred on the following shapes:

a



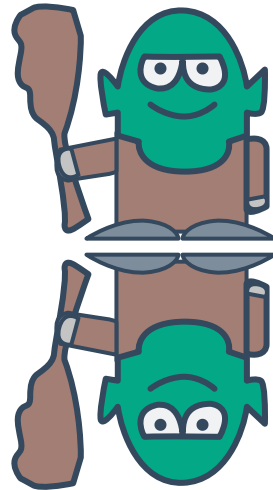
b



c



d



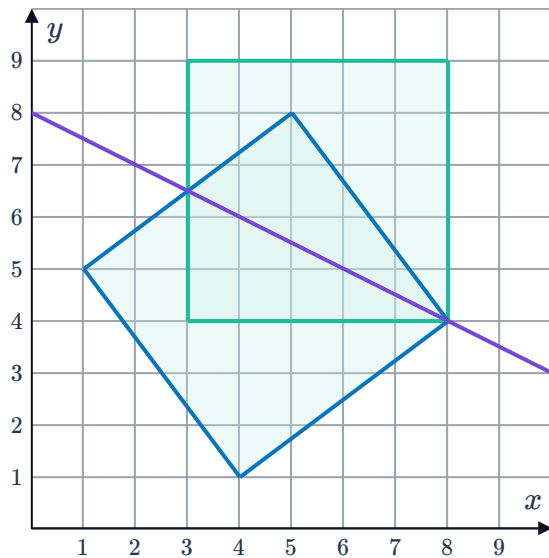
14 True or false:

A shape is reflected across a line. If the reflected shape is then reflected across the same line again, it will return to the original shape.

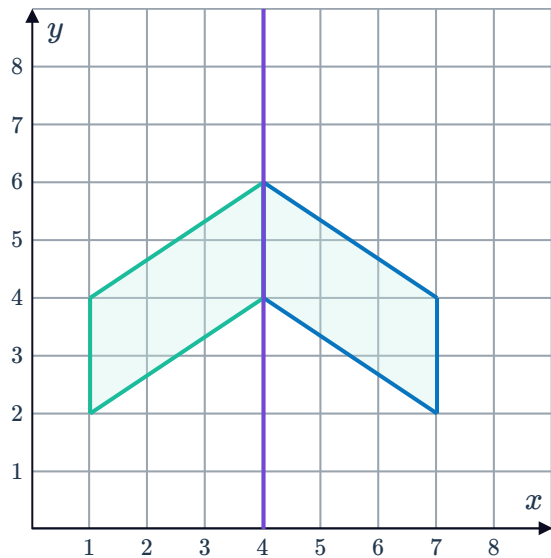
15 Determine whether the following diagram shows a reflection across the given line:



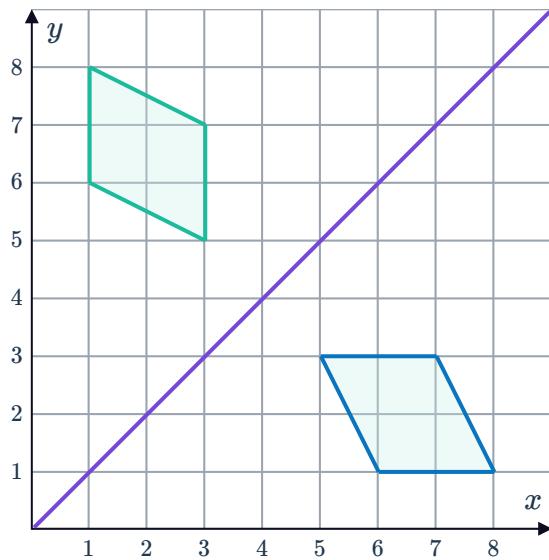
a



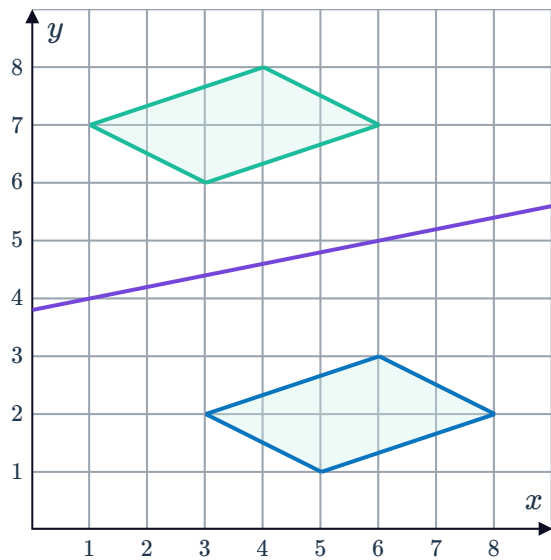
b



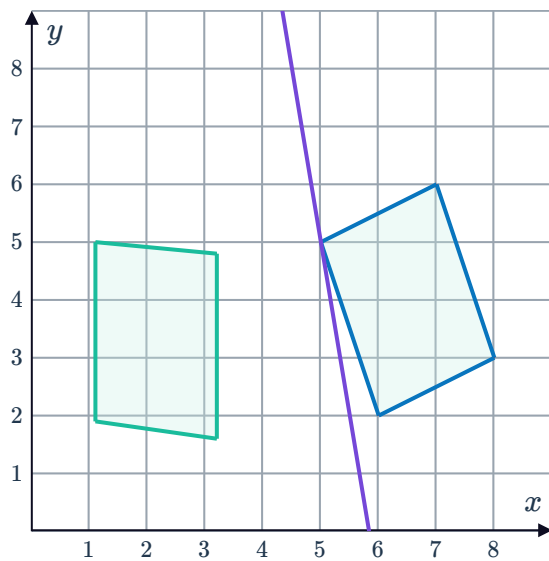
c



d



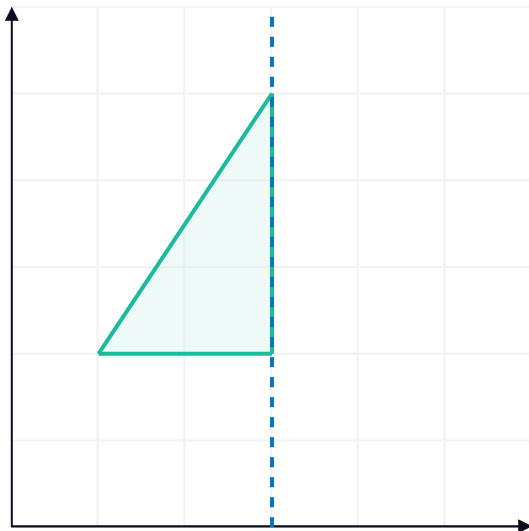
e



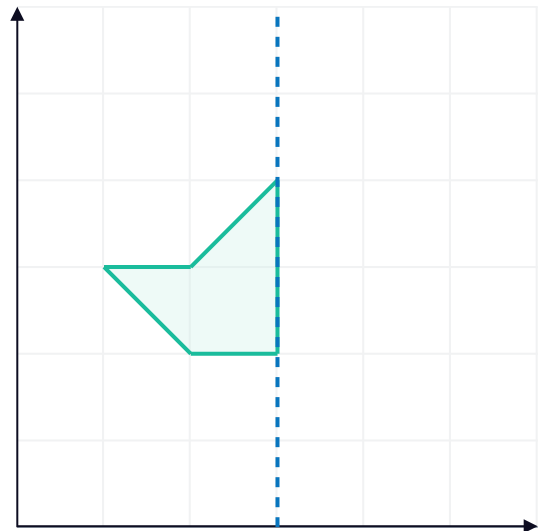
16 Reflect the following shapes about the given line:



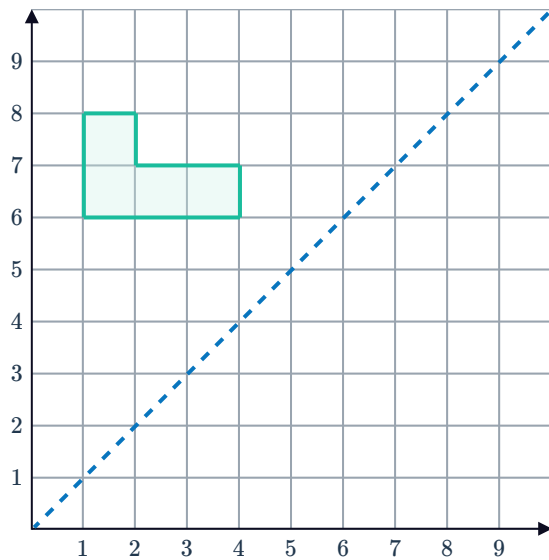
a



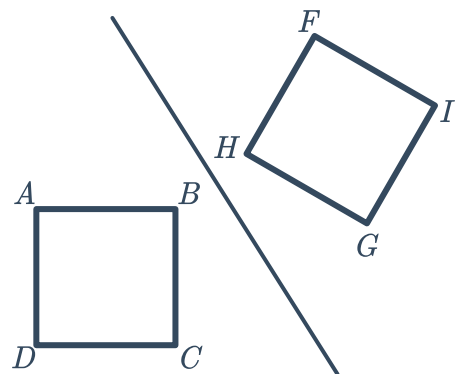
b



c



17 The square below is reflected about the given line. Which point would correspond to point A ?



18 For each of the following:



- Plot the point A on a number plane.
- Plot point A' , a reflection of point A across the indicated axis.

a $A(8, 5)$; x -axis

b $A(-10, -6)$; x -axis

c $A(-5, -3)$; y -axis

d $A(8, -6)$; y -axis

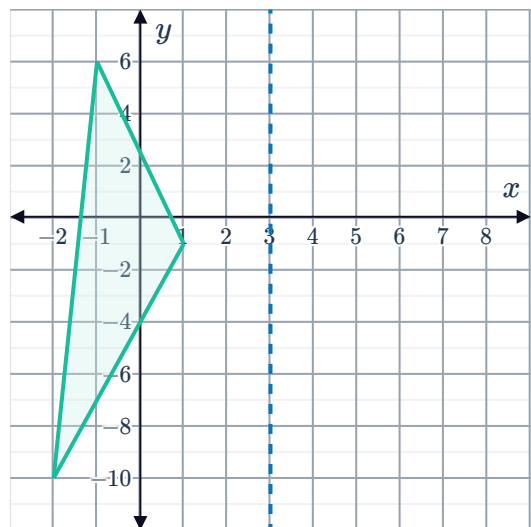
19 Consider the point $A(5, 6)$.

- a Plot point A on a number plane.
- b If point A is reflected across the y -axis to produce point B , what will be the coordinates of point B ?
- c What distance would the original point A need to be translated to overlap point B ?

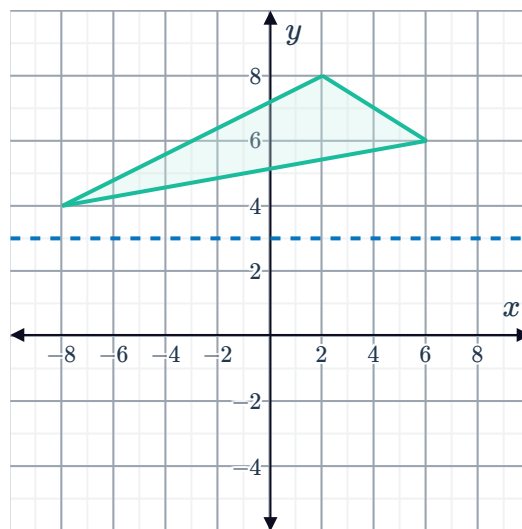
20 Consider the line segment AB , where the endpoints are $A(-9, -10)$ and $B(4, 3)$.

- a Plot the line segment AB on a number plane.
- b Plot the reflection of the line segment AB across the x -axis.

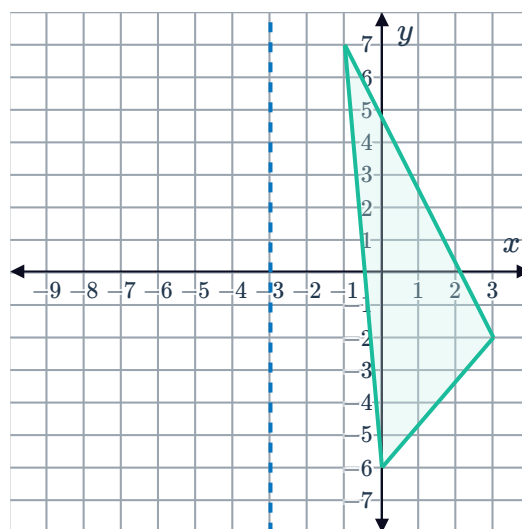
21 Plot the reflection of the given triangle across the line $x = 3$:



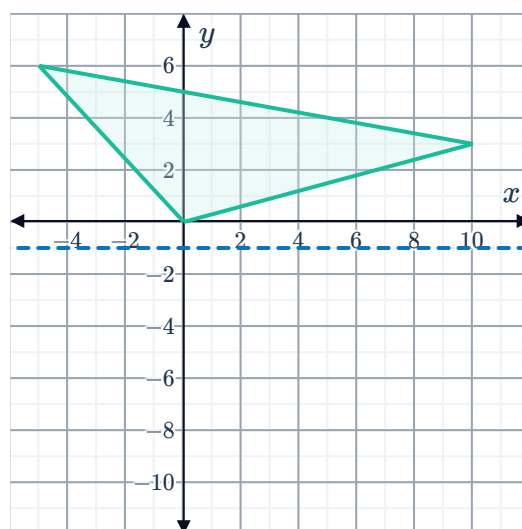
- 22 Plot the reflection of the given triangle across the line $y = 3$:



- 23 Plot the reflection of the given triangle across the line $x = -3$:



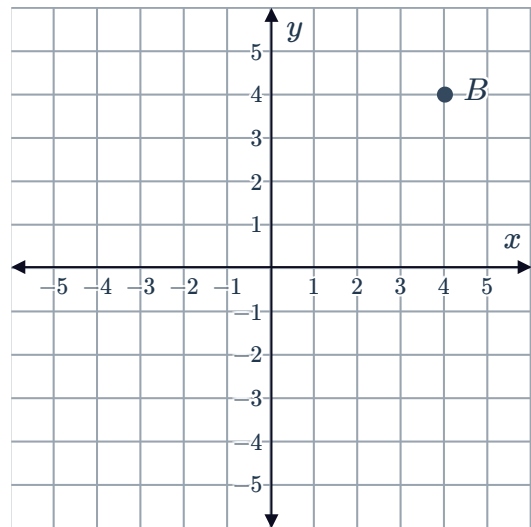
- 24 Plot the reflection of the given triangle across the line $y = -1$:





Rotations

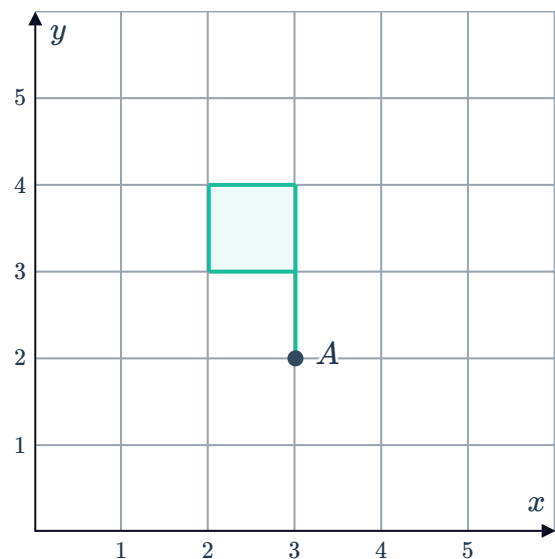
- 25** Find the coordinates of point B when it is rotated 90° clockwise about the origin.



- 26** Consider the flag on the number plane shown:

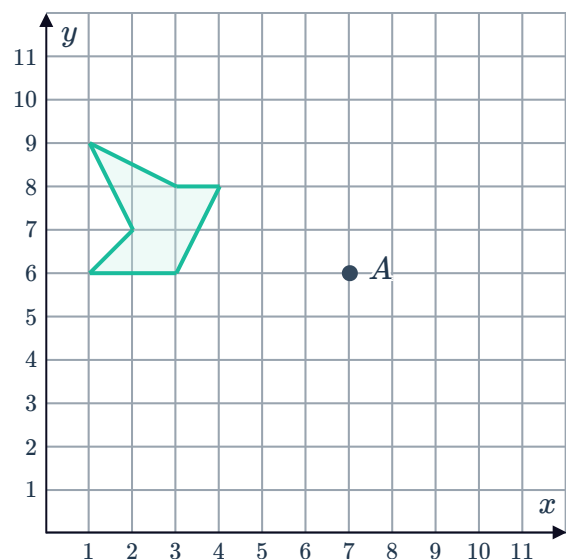
Draw the flag after it has been rotated:

- a** 90° about point A .
- b** 180° about point A .
- c** 270° about point A .



- 27** Consider the following shape:

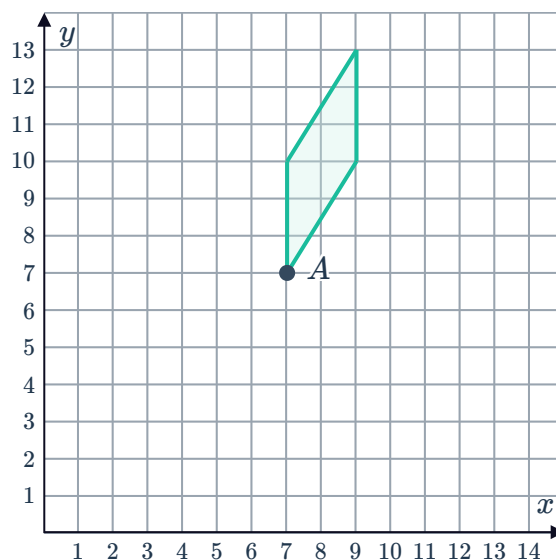
Draw the shape after it has been rotated by 180° clockwise about point A .



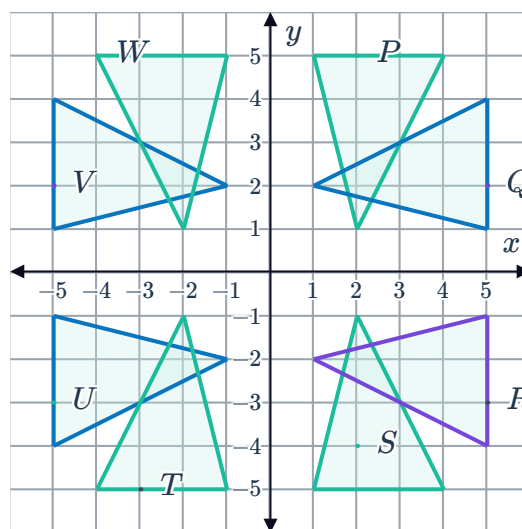
28 Consider the shape shown:



Draw the shape after it has been 135° anticlockwise rotated about point A .



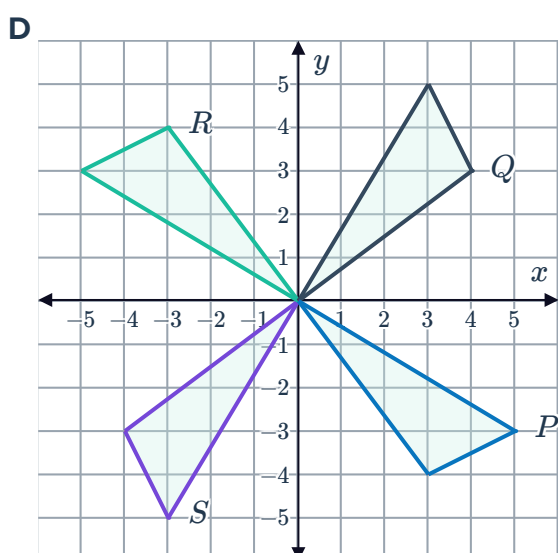
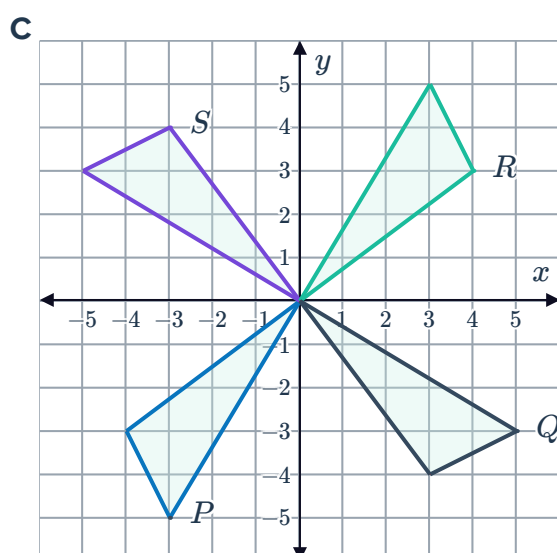
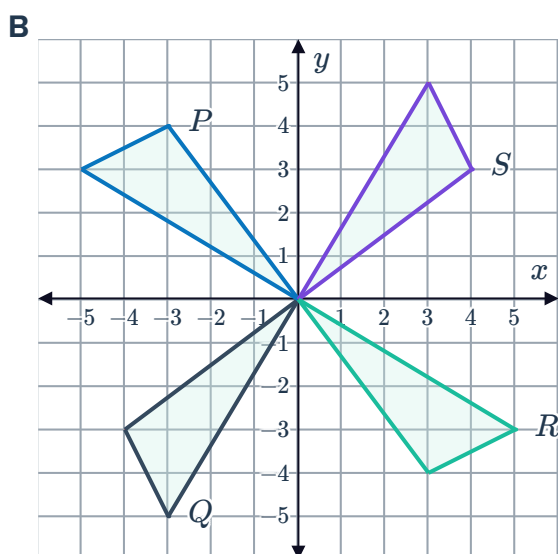
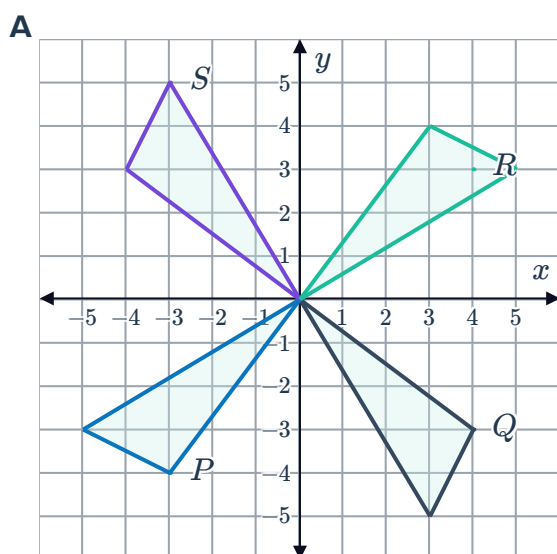
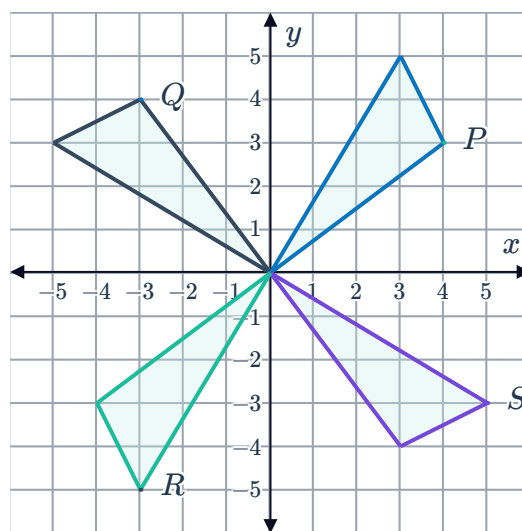
29 Which shape corresponds to a rotation of triangle R by 270° clockwise about the origin?



30 Suppose all four triangles in the image below are rotated 270° clockwise about the origin.



Which image would correspond to the result of this transformation?

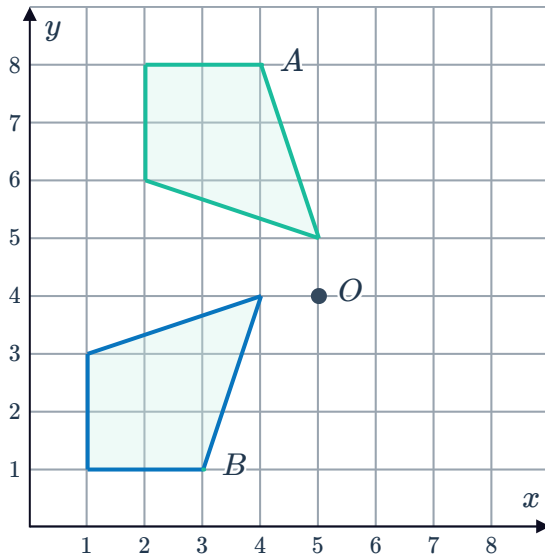


31 For each of the following:

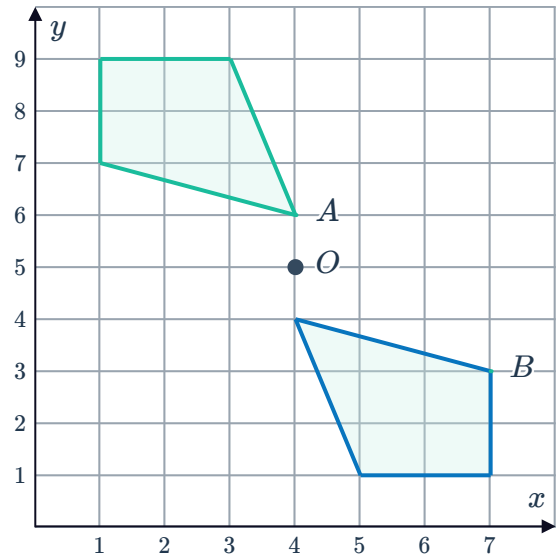


- i Find the clockwise angle of rotation of shape A to shape B about point O .
- ii Find the anticlockwise angle of rotation of shape A to shape B about point O .

a



b



32 Consider the following image:

- a Which segment represents a clockwise rotation of segment PQ by 90° about point O ?
- b Which segment represents an anticlockwise rotation of segment XY by 270° about point O ?

